

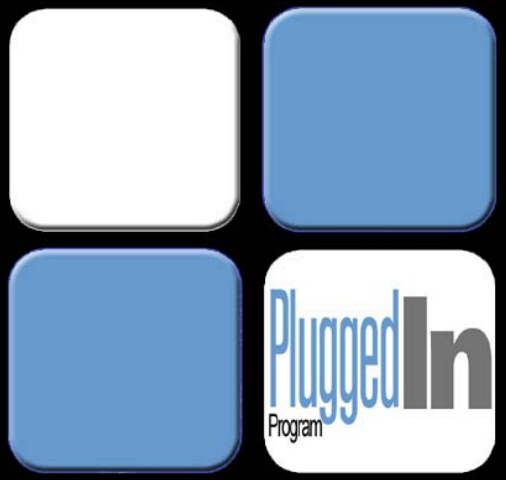
Developing for Palm's Dual Expansion Architecture

Stephen McDonnell

Business Development Manager

Agenda

- Palm™ Handheld's Expansion Strategy
- Technical Overview
- Technical Details
- Wrap-up
- Q and A



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Palm's Expansion Strategy

What is Palm's Dual Expansion Architecture



Universal Connector



SD
&
MultiMediaCard
Media



SDIO Cards



Dual Expansion Architecture

- Palm Universal Connector (PUC) system
  - Create one product for Palm's future product family
  - Enables plug & play convenience
  - Can work in concert with SD or MultiMediaCard media
  - USB capable







Dual Expansion Architecture

- Expansion card slot
  - Supports SD, MultiMediaCard and SDIO
  - Interoperable between SD enabled products
  - SDIO combination card: IO function + memory
  - Industry standard backed by over 500 combined companies





SDIO

- Expandable SD Card
  - Alleviates slot contention for memory and I/O users
  - I/O only possible for cost-sensitive users
  - Full adherence to SDA specifications for I/O and combination card functionality
  - Maximum flexibility for users with multiple I/O needs





Broad, Brand-name Support
Together, Over 500 Members

Design Decision Factors

- PUC or Expansion Card Slot …What's the difference?
  - Size
  - Speed
  - Power
  - Feasibility
  - Ease of development



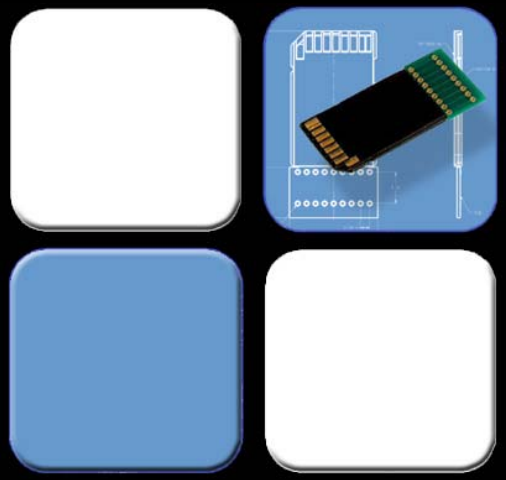
Why a Dual Expansion Architecture?

- Not limited to one expansion option
  - GPS receiver and maps
  - MP3 player and music
- Different I/O requires different form factors
  - Large sizes
  - More power
  - Faster transfer
- Need to support legacy devices
  - Sled design is well known



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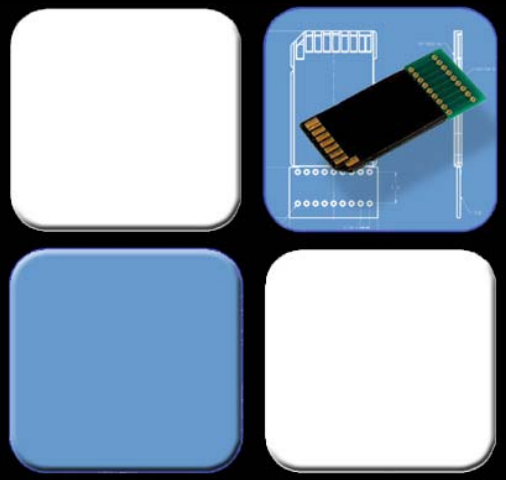




Technical Overview

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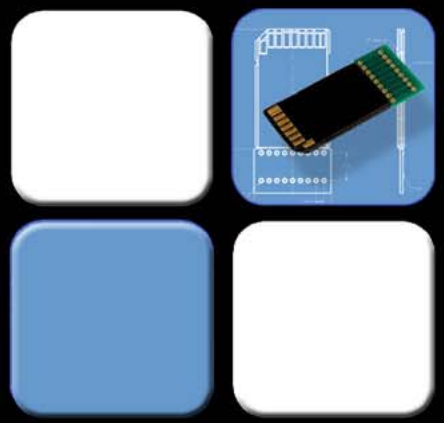




Palm Universal Connector System Technical Overview

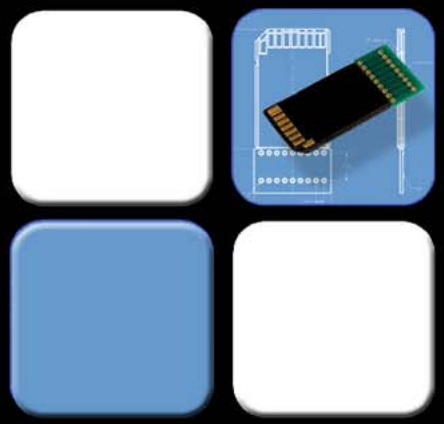
Universal Connector Overview

- Palm standard connector for 2-3 years
  - Introduced with the m500 series
  - Stable market
- Supports traditional sled designs
- Lots of room
- Easy to create
- Lower speed
- Connectors and plastic available from Palm™ Expansion Parts Store



Universal Connector Overview

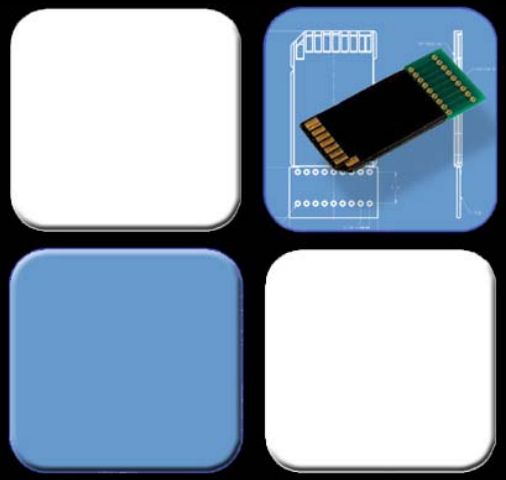
- Serial I/O
- USB possible, but only as slave
- Power supplied (3.2V at <100 mA)
- Pass-thru charging supported
- Interrupt supported
- Plug and Play (PnP) supported





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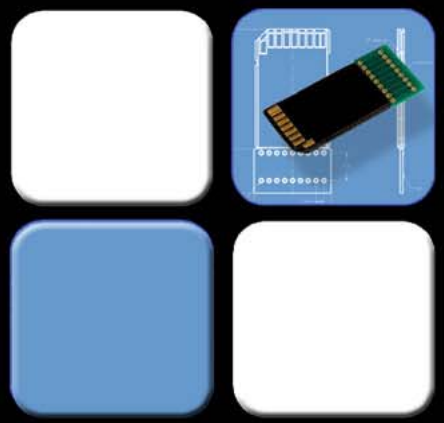




Expansion Slot & Card Technical Overview

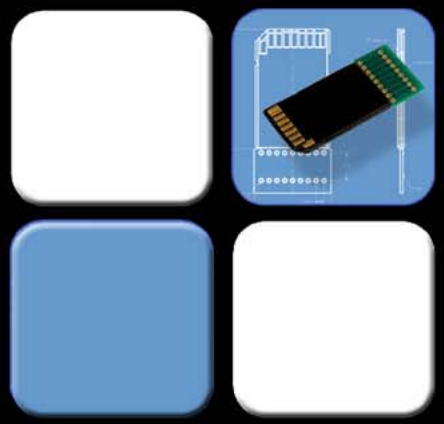
Expansion Slot & Card Overview

- Versatile
  - One slot will accept:
    - MultiMediaCard
    - SD Card
    - SDIO Card
- New form factor
  - Postage Stamp sized
  - Light
  - Rugged
- Fast
  - Currently 1MB/sec
  - Capable of up to 10 MB/sec



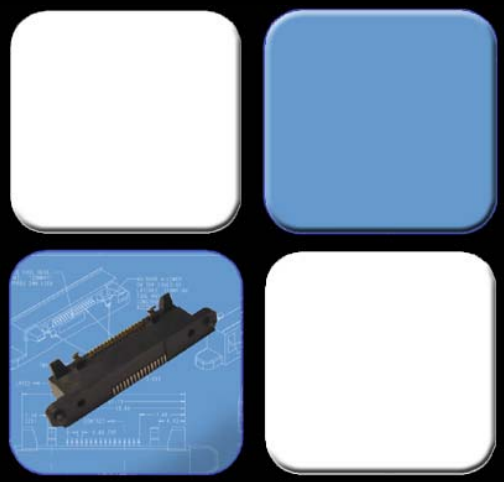
Expansion Slot & Card Overview

- Based on open industry standards
  - MMCA (MultiMediaCard Association)
    - 100+ members
    - www.mmca.org
  - SDA (SD Association)
    - 400+ members
    - www.sdcard.org
- Designed for mobile devices
  - Small
  - Light
  - Rugged
  - Low power



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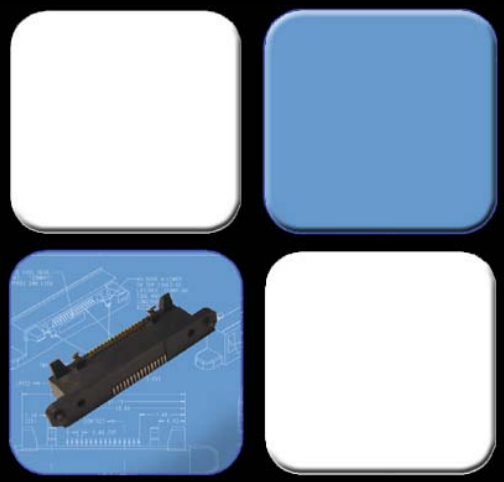




Technical Details

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Palm™ Universal Connector System Technical Details

Universal Connector

Detailed Look at the 16 pins

- 2 pins for charge
  - 5V at 1A DC
  - Can be used for pass-thru charge of sled battery
  - No connect for primary battery devices (m125)
- 5 pins for RS-232
  - Same pins as used for prior connectors
  - Speed boosted to 230 KBaud
- 3 pins for USB
  - Full speed device (12 Mb/sec)
  - Slave only
  - Self powered device



Universal Connector

Detailed Look at the 16 pins

- 2 pins for power to attached device
  - Currently 3.2V
  - Absolute max power of 100 mA!
  - Hot Plug
- 1 pin for interrupt (Hotsync™/IRQ)
  - Cradle devices auto-route as Hotsync® request
  - Peripherals route as user-defined IRQ
- 1 pin for device ID
  - Fixed resistor to ground identifies general device type/protocol
  - Used to set up communications (RS-232/USB)



Universal Connector

Detailed Look at the 16 pins

- 1 pin for attach/detach detect
  - Generates a separate IRQ on attach/detach
  - Program can register for notification
  - Identified as GND in documentation
- 1 pin spare (no connection)
  - Future expansion, don't connect on current peripherals!



Universal Connector
PnP Operation

- Attach/Detach detected via hardware
- Initial device ID based on resistor in device
  - USB or RS-232?
  - Cradle or peripheral?
- If RS-232 peripheral attached, start PnP
  - Handheld waits for PnP string from peripheral
  - PnP peripheral sends 18 bytes of data at 9600 Baud containing:
    - Vendor ID (issued by Palm, Inc.)
    - Device ID (created by vendor)
    - Maximum serial data rate
    - Links to programs and file system
    - Control Flags



Universal Connector
PnP Operation (continued)

- Once ID string has been read, the Palm will:
  - Set the Baud rate to the max value of the peripheral
  - Check the flags for programs or a file system
  - Load any programs
  - Use the program's flags to:
    - Auto-launch the program
    - Make the program "sticky"
    - Auto-reset after load
    - Set the no overwrite mode
  - Look for and mount any file system
    - FAT12/16 based
    - Read and Write possible



Universal Connector

PnP Wrap-up

- Vendor unique command support
  - A simple protocol to read and write data
  - Optional for developers
- PnP spec is extensible
  - Room for future upgrades
- More information including the developer documents are available at the PluggedIn Program web site:
  - www.palm.com/pluggedin





Stephen McDonnell

Demo of Xircom 802.11b sled

Bill Stanley



Design Example

- Multiple methods to implement PnP
- Use a microprocessor
  - Almost free if the peripheral already has a microprocessor
  - Add a very low-cost microprocessor
    - PIC
    - 8051
- Use hardware logic
  - Built into peripheral hardware
  - Small PLD
- Additional information is available on the PluggedIn Program web site



Things to watch out for!
(Electrical Design)

- Power drain
  - Ideally 20 mA
  - External batteries
- Shutdown modes
- RS-232 driver
- Forcing power backwards
- Input capacitance



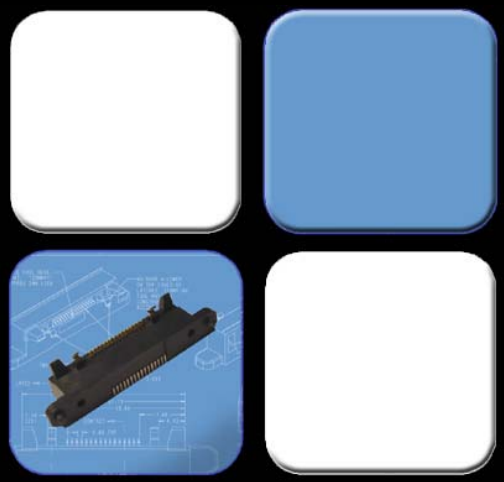
Things to watch out for!
(Mechanical Design)

- The Palm Universal Connector (PUC) is a system
- Use the guidelines!
- Make it as thin as possible
- Mechanical design
- Look at available plastic and latches
- Talk to Palm!



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Expansion Slot & Card Technical Details

SDIO
Detailed Look

- SDIO is based on the SD memory form factor
  - Plugs into any SD socket
  - Extension allows larger “bumps”
  - Added locking notches
- Multi-function cards are supported
  - 1-7 functions per card
  - plus SD memory (combo card)
- Many concepts based on PCMCIA I/O cards
  - PnP designed in from the start
  - Uses extensible CIS
    - Vendor ID’s come from PCMCIA



SDIO
Detailed Look (continued)

- Interrupts supported
  - Special case in 4-Bit mode
- 3 new commands created
  - SDIO Initialize
  - R/W single byte
    - Read after write option
  - R/W multiple
    - Multi-byte
    - Multi-block



SDIO
Detailed Look (continued)

- Low power
  - Less than 200 mA peak
  - Lower power at insert
  - Initialization Voltage: 2.0 to 3.6V
  - Operational Voltage range: 3.1 to 3.5V
- New optional protocols:
  - Read wait
    - Provides for wait states
  - Suspend/Resume
    - A simple multi-tasking ability



SDIO
Detailed Look (continued)

- Support for code storage in the card
  - Up to 16 MB of code storage (R/O or R/W)
  - Drivers and/or applications
    - Fat12/16
    - Mounts as VFS
- 2 classes of SDIO cards are defined
  - Low Speed
    - SPI and 1-bit modes are mandatory
    - 400 KHz is the only guaranteed clock
  - Full Speed
    - SPI, 1-bit and 4-bit modes are mandatory
    - Clock support up to 25 MHz required



Stephen McDonnell

Palm SDIO Bluetooth Demo



Bill Stanley

Getting Started with SDIO

- First step… Contact the SDA
  - www.sdcard.org
  - Join and sign the NDA to see the specifications
- If you want to develop SDIO cards:
  - Sign the Host/Ancillary Product License Agreement (HALA)
  - If your design has more than 2 MB of CSA, obtain and sign the the Card License Agreement (CLA)
    - Note: there may be additional royalties



Getting Started with SDIO
(continued)

- Second step… Connect with Palm
- Join the PluggedIn Program
  - www.palm.com/pluggedin
- PluggedIn Program resources include:
  - Design guides
  - White papers
  - Plastics
  - CAD tools and models
  - Extender cards
  - Other design aids



Getting Started with SDIO
(continued)

- Look at the SDIO EDK available from the Palm Expansion Parts Store
  - Kit contains a μP based reference board
  - Temperature sensor with additional pins
- All documentation supplied
  - Schematic
  - Code listing
  - PCB Layout
  - PLD source
  - Sample driver App



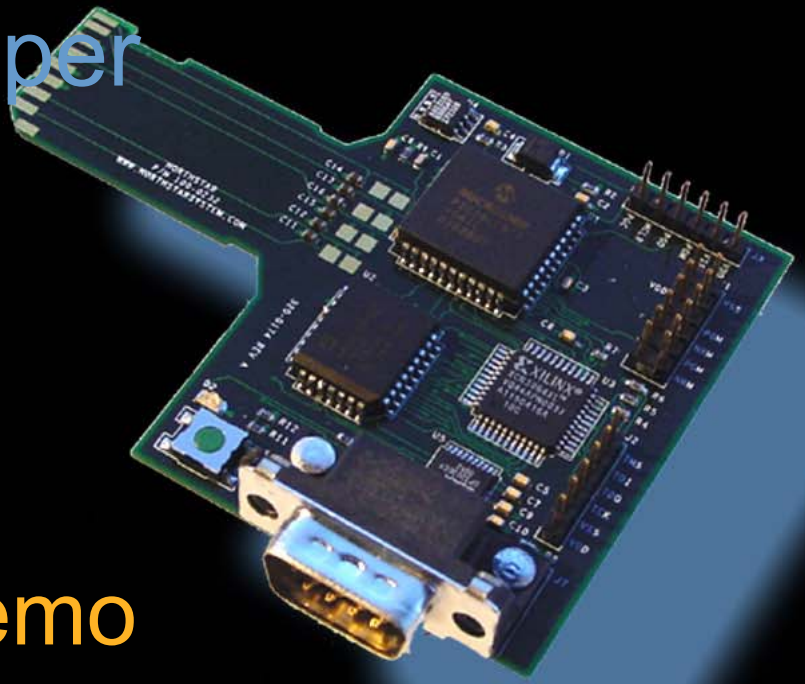
Getting Started with SDIO
(continued)

- SDIO Development Contest
  - To be announced at PalmSource®
  - PluggedIn Program members to be notified with more information soon





Paul D. Leeper



Palm EDK Board Demo

Bill Stanley

Interfacing Existing I/O Functions with the SD Bus?

- Design details
  - How fast is the I/O device?
    - If greater than 40KB/sec, high speed is mandatory
      - Must use interface hardware (chip)
      - Must support 100 Mb/sec transfer
        - Camera
    - If less than 40KB/sec, use low speed
      - Modems
      - Human input devices
      - GPS



Interfacing Existing I/O Functions with the SD Bus?
(continued)

- Low speed…Microprocessor or chip?
  - A μP can handle the low speed communications between the function and the host
    - Higher cost
    - Greater effort (H/W and S/W)
    - May require some hardware assist
      - Shift Register



Software

- Slot driver
- API's
- White paper





Things to Watch Out For

- Fragile case points
- Location on Palm™ handhelds
  - m505 versus m125
- Don’t overhang the top or bottom of the Palm
  - Extreme stress on card possible
- Watch the current! (operation and inrush)
  - Peak max current is 200 mA
  - Devices should draw less than 100 mA to be battery friendly
  - Limit input capacitance to less than 7.5 µF
  - Look at inrush when enabling subsystems



Things to Watch Out For
(continued)

- If the card is low speed it must:
  - Support both SPI and 1-bit modes
  - Function if the max clock rate is 400 KHz
- If the card is full speed it must:
  - Support SPI, 1-bit and 4-bit modes
  - Operate from 0-25MHz
- Low power mode support is important
- Watch the size of the CSA
  - More than 2MB may require royalties



Things to Watch Out For
(continued)

- Make sure card operates at initialization voltage
  - The card must communicate at 2.0-3.6V
  - The card must operate at 3.1-3.5V
- The time between packets is small
  - Make sure the card can keep up
- Keep Palm informed of your plans

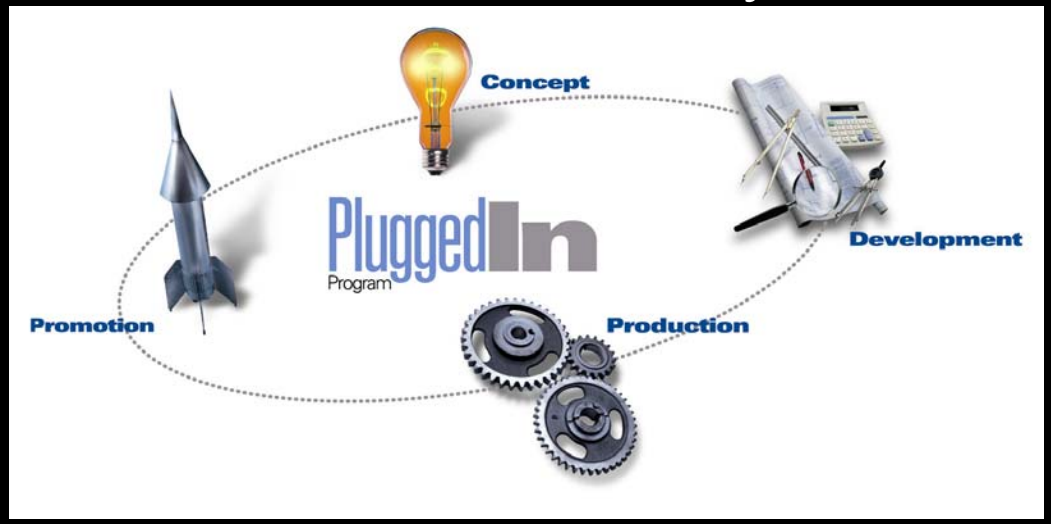


Plugged-in Program

www.palm.com/pluggedin

- Provides the resources you need to develop expansion solutions
- New and improved website

- 3 membership levels
  - Member
  - Marketing
  - Business
- For expansion developers only

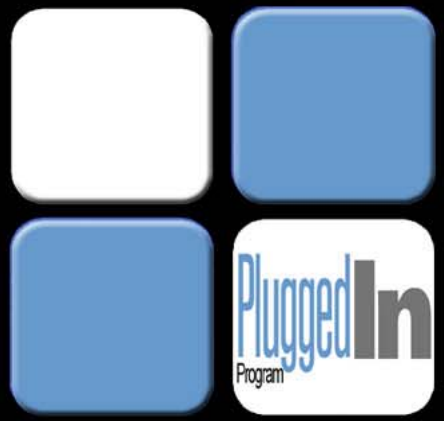


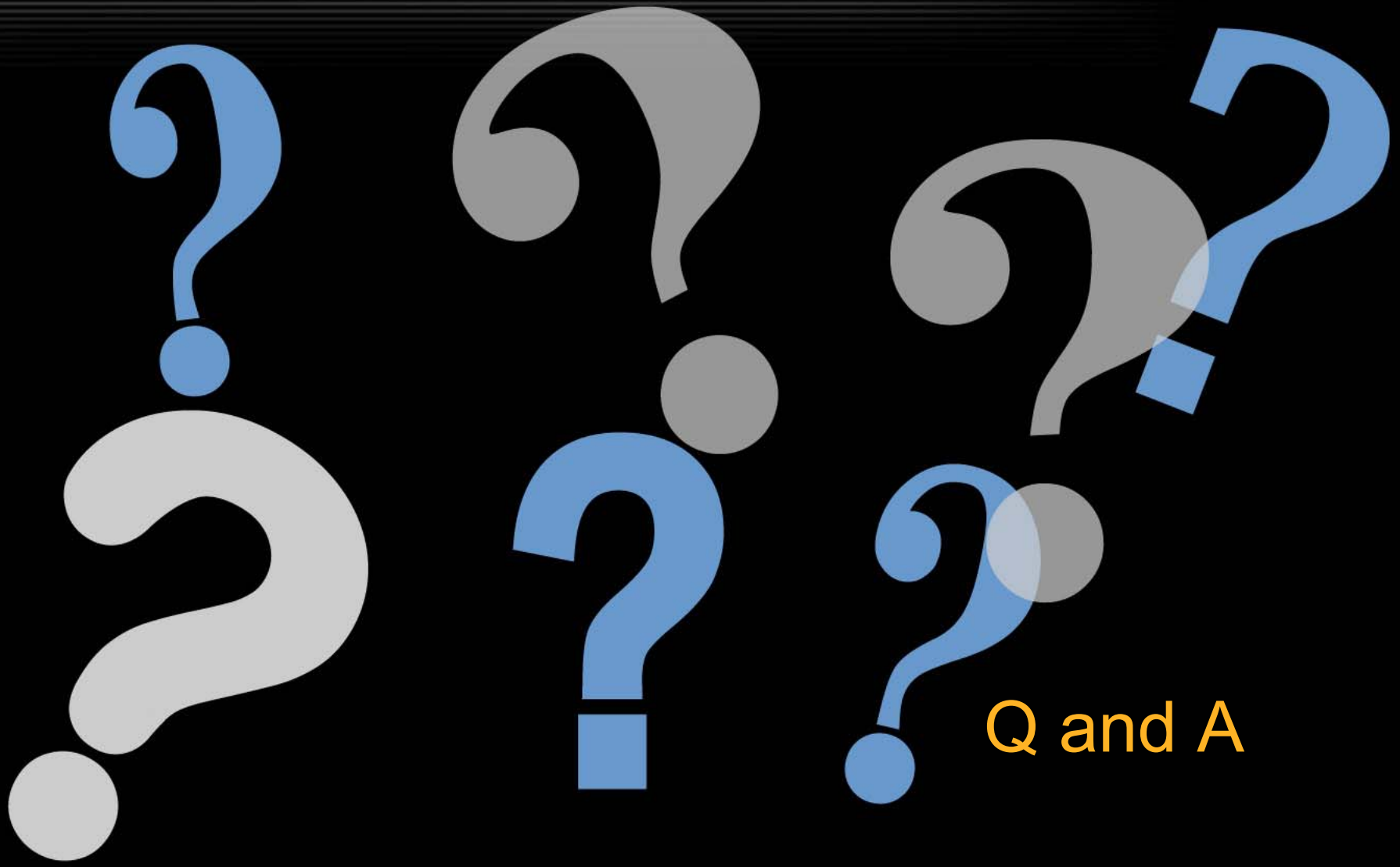


Wrap-up

Wrap-up

- Palm's Expansion Strategy
  - Palm Universal Connector (PUC) system
  - Expansion Card Slot
- Technical Information
  - Palm™ Universal Connector (PUC) system
  - Expansion Card Slot
  - SDIO
- Resources
  - PluggedIn Program
  - SD Association





Q and A


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